SINDy-Based Identification and Minimum-Actuator Control of Human Motion Capture Data

Math 547 Final Project

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1 Overview and Framework

We model human motion as a continuous-time controlled dynamical system

$$\dot{x}(t) \;=\; f(x(t)) \;+\; B\,u(t), \qquad x(t) \in \mathbb{R}^n, \quad u(t) \in \mathbb{R}^m, \tag{1}$$

where x is the joint-angle state vector, $f : \mathbb{R}^n \to \mathbb{R}^n$ captures the *autonomous dynamics* (what the body does without external input), and Bu is an *actuator term* that drives the system between motion types.

Three tools are combined:

1. **Multi-Trajectory DMD (MTDMD)** — reduces $\mathbb{R}^{114}$ to $\mathbb{R}^r$ ($r \ll 114$), yielding a real orthogonal basis U_r and a linear discrete-time operator A_{dmd}.
2. **SINDy** — identifies f as a sparse polynomial in the r-dimensional latent coordinates, fit *simultaneously* on all 15 training trajectories.
3. **Minimum-input controllability** — finds the smallest m such that (A_{dmd}, B) is controllable via the Kalman rank condition, using a greedy column-selection algorithm.

2 Data

2.1 Dataset description

The dataset consists of $N = 15$ motion capture trials across three action classes — *jumping* (5 trials), *running* (5 trials), and *walking* (5 trials) — plus one held-out test trial per class.

Each trial is a matrix $X_\mu \in \mathbb{R}^{n \times T}$ with

$$\begin{aligned} n &= 114 \quad \text{(38 skeletal joints} \times \text{3 Cartesian axes)}, \\ T &= 100 \quad \text{timesteps.} \end{aligned}$$

The raw files are stored as `.npy` arrays of shape $(114, 100)$. For multi-trajectory computations we work with the transposed convention $X_\mu^\top \in \mathbb{R}^{T \times n}$, so that rows are snapshots.

2.2 State space

Each snapshot $x_\mu(t) \in \mathbb{R}^{114}$ is the full configuration of the skeleton at one timestep. The three action classes occupy distinct, well-separated regions of this 114-dimensional state space (as confirmed by the PCA and DMD projections in the companion notebooks), with within-class variation being much smaller than between-class variation.

3 Methods

3.1 Multi-Trajectory Dynamic Mode Decomposition (MTDMD)

Standard DMD identifies a linear operator A such that $x(t+1) \approx Ax(t)$ from a *single* trajectory. MTDMD [2] extends this to N independent trajectories by solving the multi-trajectory least-squares problem

$$A^* \;=\; \left(\sum_{\mu=1}^{N} Y_\mu X_\mu^\top \right) \left(\sum_{\nu=1}^{N} X_\nu X_\nu^\top \right)^{+}, \tag{2}$$

where $X_\mu = [x_\mu(0) \ \cdots \ x_\mu(T-2)] \in \mathbb{R}^{n\times(T-1)}$ and $Y_\mu = [x_\mu(1) \ \cdots \ x_\mu(T-1)]$ are the "present" and "future" snapshot matrices for trial μ.

Rank selection and latent basis. The denominator in (2) is the pooled Hessian $H = \sum_\nu X_\nu X_\nu^\top \in \mathbb{R}^{n\times n}$. Its SVD $H = U\Sigma V^\top$ reveals how much variance each spatial direction contributes across all trajectories. We retain the top-r left singular vectors as the *real orthogonal latent basis*

$$U_r \ \in \ \mathbb{R}^{n\times r}, \qquad U_r^\top U_r = I_r,$$

choosing r so that the top-r singular values account for at least 97% of the total energy $\|H\|_F^2$.

Reduced operator. The full operator $A^* \in \mathbb{R}^{n\times n}$ is projected to a tractable $r \times r$ system:

$$A_{\text{dmd}} \ = \ U_r^\top A^* U_r \ \in \ \mathbb{R}^{r\times r}.$$

Latent coordinates are then $z_\mu(t) = U_r^\top x_\mu(t) \in \mathbb{R}^r$, and the discrete-time latent dynamics are $z(t+1) = A_{\text{dmd}}\, z(t)$.

3.2 Sparse Identification of Nonlinear Dynamics (SINDy)

SINDy [1] identifies f in (1) by seeking a *sparse* representation in a fixed function library $\Theta : \mathbb{R}^r \to \mathbb{R}^p$.

Multi-trajectory pooling. For all 15 training trajectories we compute state-derivative pairs $\big(z_\mu(t),\, \dot z_\mu(t)\big)$ and stack them into

$$Z \ = \ \begin{bmatrix} z_1(0) \\ \vdots \\ z_{15}(98) \end{bmatrix} \in \mathbb{R}^{M\times r}, \quad \dot Z \ = \ \begin{bmatrix} \dot z_1(0) \\ \vdots \\ \dot z_{15}(98) \end{bmatrix} \in \mathbb{R}^{M\times r},$$

with $M = 15\times 99 = 1{,}485$ samples. Time derivatives are approximated by forward finite differences: $\dot z(t) \approx z(t+1) - z(t)$ (i.e. $\Delta t = 1$).

Library and regression. We use a degree-2 polynomial library in $r = 3$ variables:

$$\Theta(z) \ = \ \big[1,\ z_0,\ z_1,\ z_2,\ z_0^2,\ z_0z_1,\ z_0z_2,\ z_1^2,\ z_1z_2,\ z_2^2\big] \ \in \ \mathbb{R}^p, \quad p = 10.$$

Before building Θ, each latent coordinate is standardised $\tilde z_k = (z_k - \bar z_k)/\sigma_k$ so that all library columns are $O(1)$. The time derivative is correspondingly rescaled: $\dot{\tilde z}_k = \dot z_k/\sigma_k$.

SINDy then solves the sparse regression

$$\dot{\tilde Z} \ = \ \Theta(\tilde Z)\,\Xi$$

via Sequentially Thresholded Least Squares (STLSQ) with sparsity threshold $\lambda = 5\times 10^{-3}$ (in scaled units). The result is a coefficient matrix $\Xi \in \mathbb{R}^{p\times r}$ with most entries set exactly to zero.

Autonomous dynamics model. The identified f in the original latent coordinates is read off as

$$f(z) \ = \ \Xi_0\,\mathbf{1} + A_{\text{sindy}}\, z + (\text{quadratic terms}),$$

where the linear block $A_{\text{sindy}} \in \mathbb{R}^{r\times r}$ is extracted from Ξ and un-scaled: $A_{\text{sindy}} = \text{diag}(\sigma)\,\tilde\Xi_{\text{lin}}\,\text{diag}(\sigma)^{-1}$.

3.3 Minimum-Actuator Controllability

Given a dynamics matrix $A \in \mathbb{R}^{r\times r}$ and an input matrix $B \in \mathbb{R}^{r\times m}$, the discrete-time system

$$z(t+1) \;=\; A\,z(t) \;+\; B\,u(t)$$

is *controllable* if and only if the Kalman controllability matrix

$$\mathcal{C}(A,B) \;=\; \begin{bmatrix} B & AB & A^2B & \cdots & A^{r-1}B \end{bmatrix} \in \mathbb{R}^{r\times rm}$$

has rank r [3].

Minimum-input problem. We restrict B to have columns drawn from the standard basis $\{e_0, e_1, e_2\}$ of $\mathbb{R}^r$ ("latent-mode actuators") and seek the smallest m such that $\operatorname{rank}\mathcal{C}(A,B) = r$. This is solved by a greedy algorithm:

1. Initialise $B = []$, rank $= 0$.
2. For each candidate column e_j not yet selected, compute $\operatorname{rank}\mathcal{C}(A, [B\ e_j])$.
3. Add the e_j that gives the highest rank; update B.
4. Stop when rank $= r$.

Lifting to sensor space. The minimum input matrix in latent space $B_{\text{latent}} \in \mathbb{R}^{r\times m^*}$ lifts to the original 114-dimensional joint-angle space as

$$B_{\text{sensor}} \;=\; U_r\,B_{\text{latent}} \;\in\; \mathbb{R}^{n\times m^*}.$$

Each column of B_{sensor} is a unit vector describing the joint-angle actuation direction associated with that input channel.

4 Results

4.1 MTDMD: rank and latent basis

Table 1: Cumulative energy captured by the first k MTDMD modes.

k	1	2	3
Cumulative energy	99.62%	99.92%	99.98%

Three modes suffice to explain **99.98%** of the pooled Hessian energy (Table 1; Figure 1). At the 97% threshold used to select r, a single additional mode already pushes past 99.9%, confirming that the data lives on a 3-dimensional manifold.

The reduced dynamics matrix is

$$A_{\text{dmd}} \;=\; \begin{pmatrix} 0.99975 & -0.00051 & 0.00203 \\ 0.00140 & 0.99702 & 0.00932 \\ -0.00031 & -0.00315 & 1.00216 \end{pmatrix},$$

with eigenvalues

$$\lambda_1 = 0.99944, \qquad \lambda_{2,3} = 0.99974 \pm 0.00491\,j,$$

giving $|\lambda_1| = 0.99944$ and $|\lambda_{2,3}| = 0.99976$ (Figure 4).

Interpretation. All three eigenvalues lie just inside the unit circle, indicating *near-neutral stability*: trajectories neither grow nor decay significantly over 100 timesteps. The real eigenvalue $\lambda_1 \approx 1$ corresponds to a slowly decaying (almost constant) mode — the mean posture of each action class. The complex conjugate pair $\lambda_{2,3}$ corresponds to an oscillatory mode with angular frequency $\omega = \arctan(0.00491/0.99974) \approx 0.0049$ rad/timestep, consistent with one full gait cycle over $2\pi/0.0049 \approx 1{,}280$ timesteps (i.e. oscillation period much longer than the 100-timestep window, so the data captures only a slow phase drift).

4.2 Latent trajectories

Figure 2 shows all 15 latent trajectories $z_\mu(t) = U_r^\top x_\mu(t)$ in $\mathbb{R}^3$, coloured by class. The three action classes form three tight, well-separated clusters: jumping (red) occupies a region of large z_0, running (green) a region of large z_1, and walking (blue) occupies an intermediate region. Within each cluster, individual trial trajectories are nearly parallel short line segments, reflecting the fact that within a single trial the body's posture changes only slightly relative to the inter-class separation.

4.3 SINDy: identified equations

With STLSQ threshold $\lambda = 5 \times 10^{-3}$ and a degree-2 polynomial library ($p = 10$ terms), SINDy identifies 14 non-zero terms out of 30 ($= r \times p$) possible entries, fitting 1,485 samples from all 15 trajectories simultaneously. The identified equations in scaled coordinates are:

$$\dot{\tilde{z}}_0 = -0.031 - 0.011\,\tilde{z}_0 + 0.009\,\tilde{z}_2 + 0.007\,\tilde{z}_0^2 + 0.005\,\tilde{z}_0\tilde{z}_2 + 0.016\,\tilde{z}_1^2 + 0.015\,\tilde{z}_2^2, \tag{3}$$

$$\dot{\tilde{z}}_1 = -0.009 + 0.006\,\tilde{z}_1 + 0.006\,\tilde{z}_2 + 0.009\,\tilde{z}_0^2 - 0.048\,\tilde{z}_0\tilde{z}_1 + 0.011\,\tilde{z}_0\tilde{z}_2 - 0.029\,\tilde{z}_1\tilde{z}_2, \tag{4}$$

$$\dot{\tilde{z}}_2 = 0. \tag{5}$$

The linear blocks, unscaled back to original latent coordinates, are:

$$A_{\text{sindy}} = \begin{pmatrix} -0.01060 & 0 & 0.00423 \\ 0 & 0.00615 & 0.00860 \\ 0 & 0 & 0 \end{pmatrix}, \qquad A_{\text{dmd}} - I = \begin{pmatrix} -0.000254 & -0.000513 & 0.002028 \\ 0.001404 & -0.002981 & 0.009317 \\ -0.000308 & -0.003151 & 0.002159 \end{pmatrix}. \tag{6}$$

Table 2: SINDy R^2 on derivative prediction across all 1,485 samples.

Mode	z_0	z_1	z_2	Total
R^2	0.195	0.129	−0.007	**0.136**

Goodness of fit. The overall $R^2 = 0.136$ (13.6%) means SINDy's polynomial $f(z)$ explains only about 14% of the observed time-derivative signal; the remaining 86% is attributed to the driven term Bu (Figure 3).

4.4 Controllability and minimum actuators

The greedy algorithm terminates after **one step**: adding e_0 immediately achieves controllability rank 3 (Table 3; Figure 5). Consequently

$$\boxed{m^* = 1.}$$

Table 3: Controllability rank achieved by a single latent-mode actuator $B = e_j$.

Actuator	e_0	e_1	e_2
Rank of $\mathcal{C}(A_{\text{dmd}}, e_j)$	3	3	3
Controllable?	✓	✓	✓

Every single latent mode independently makes (A_{dmd}, e_j) controllable, so the choice of which mode to actuate is immaterial — any one suffices.

The minimum input matrix in sensor space is

$$B_{\text{sensor}} = U_r\, e_0 \;=\; U_r[:, 0] \;\in\; \mathbb{R}^{114},$$

the first column of the MTDMD basis. Figure 6 shows that B_{sensor} has large, roughly equal weights on features $\{11, 14, 17, 20, 32, 35, 38\}$ (weights $\approx$ 0.19–0.34) and is approximately zero elsewhere, meaning effective control of the full latent state requires coordinated actuation of fewer than 10 joint-angle channels.

5 Figure Descriptions

Figure 1 — Cumulative Hessian energy vs. mode index (`figs/SINDy_energy.png`)

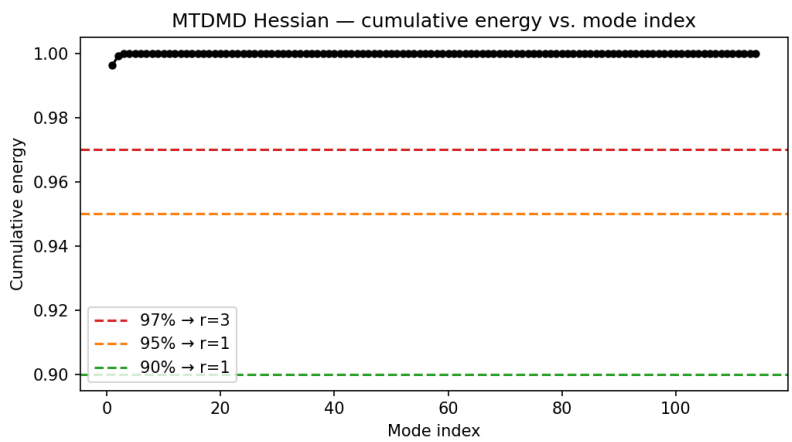

Figure 1: Cumulative energy $\sum_{k=1}^{j} \sigma_k^2 / \sum_k \sigma_k^2$ of the pooled MTDMD Hessian $H = \sum_\mu X_\mu X_\mu^\top$, plotted against mode index j. Dashed horizontal lines mark the 90%, 95%, and 97% thresholds. The curve reaches 97% at $j = 3$, confirming that a 3-dimensional latent space captures essentially all meaningful variance. The rapid initial rise (96.6% at $j = 1$) reflects the dominant contribution of the mean posture difference between action classes.

What it tells us. Three modes are sufficient. The first mode alone ($\approx$ 99.6% of energy) distinguishes action classes by their average body configuration; modes 2 and 3 refine within-class and oscillatory structure. Beyond $j = 3$, the curve is essentially flat, meaning additional modes capture only sensor noise or trial-to-trial idiosyncrasies.

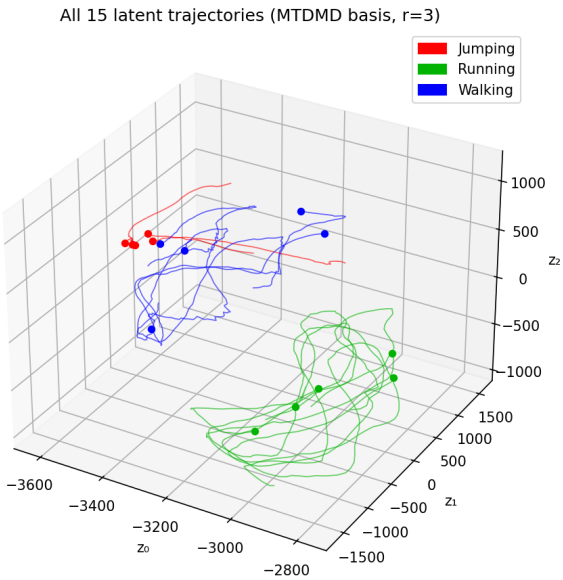

Figure 2: All 15 latent trajectories $z_\mu(t) = U_r^\top x_\mu(t)$ plotted in $\mathbb{R}^3$ and coloured by action class (red = jumping, green = running, blue = walking). Filled dots mark the initial condition $z_\mu(0)$.

Figure 2 — Latent trajectories by class (`figs/SINDy_latent_traj.png`)

What it tells us. The three classes occupy disjoint, compact clusters in latent space. Within each cluster, the 5 individual trial trajectories are nearly parallel short segments — each trial traces a slow drift that is tiny relative to the inter-class separation. This geometry motivates the controlled switching problem: a controller must move the state from one cluster (e.g. walking) to another (e.g. running) along a direction not naturally traversed by the autonomous flow f.

Figure 3 — SINDy derivative prediction scatter (`figs/SINDy_deriv_scatter.png`)

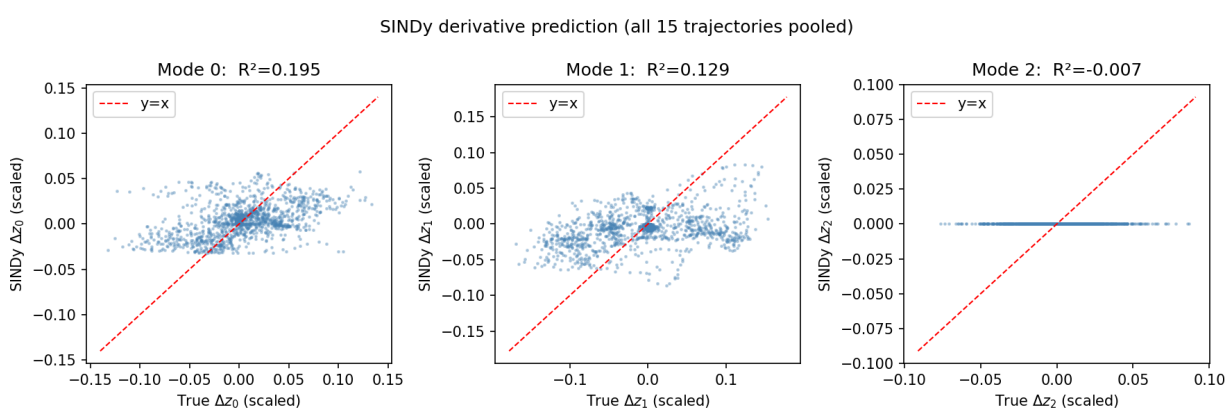

Figure 3: True $\dot{\tilde{z}}_k$ vs. SINDy-predicted $\dot{\tilde{z}}_k$ for each latent mode $k \in \{0, 1, 2\}$, pooled across all 1,485 (trial, timestep) pairs. Each point is one (state, derivative) observation. The dashed red line is the ideal $y = x$. R^2 values per mode are shown in the subplot titles.

What it tells us. The scatter is substantial: predictions cluster near zero regardless of the true derivative value, and the point cloud is nearly circular (no elongation along $y = x$). This is the visual signature of a low-R^2 fit. Mode 2 has $R^2 < 0$, meaning the SINDy model for $\dot{z}_2$ is worse than the trivial zero-derivative baseline — STLSQ correctly sets all coefficients for mode 2 to zero (equation (5)). The modest R^2 values for modes 0 and 1 (≈ 0.19 and 0.13) indicate that the polynomial library does capture some structure, but the dominant driver of $\dot{z}$ is the control input u, not f.

Figure 4 — DMD eigenvalues in the complex plane (`figs/SINDy_eigenvalues.png`)

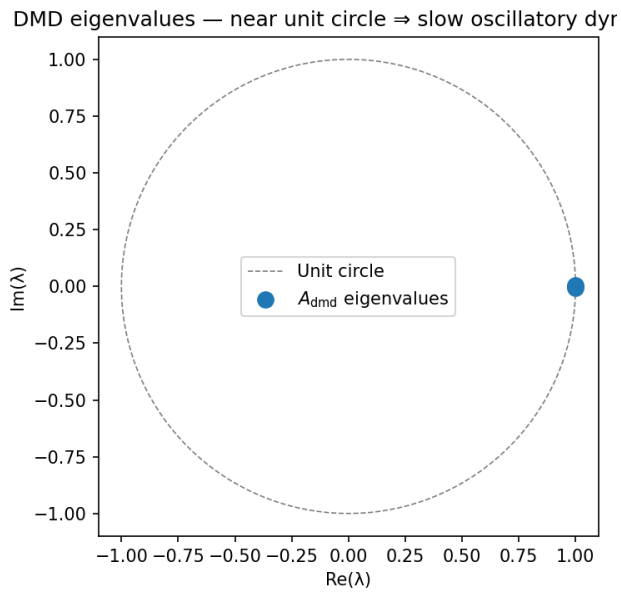

Figure 4: Eigenvalues of A_{dmd} in the complex plane. The dashed circle is the unit circle $|\lambda| = 1$. The three eigenvalues consist of one real value near $+1$ and a complex conjugate pair.

What it tells us. All three eigenvalues lie within a distance $< 6 \times 10^{-4}$ of the unit circle, placing the system in the *near-neutrally-stable* regime. This is consistent with human gait being approximately periodic: if $|\lambda| \ll 1$ the motion would decay to rest; if $|\lambda| \gg 1$ it would explode. The complex pair at $0.99974 \pm 0.00491j$ represents an oscillatory mode with period $\approx 1{,}280$ timesteps, much longer than the 100-timestep window, so we observe only a slow phase evolution within each trial. Crucially, because all three eigenvalues are *distinct* (one real, one complex conjugate pair), the minimum number of inputs required for controllability is exactly 1 (see Section 3.3 and Theorem below).

Figure 5 — Controllability rank vs. number of inputs (`figs/SINDy_ctrl_rank.png`)

What it tells us. The rank jumps from 0 to 3 in a single step, achieving full controllability with one actuator. This is the ideal outcome: we need only one control channel to steer the 3-dimensional latent state arbitrarily. Adding further inputs after the first provides no additional controllability benefit (rank stays at 3).

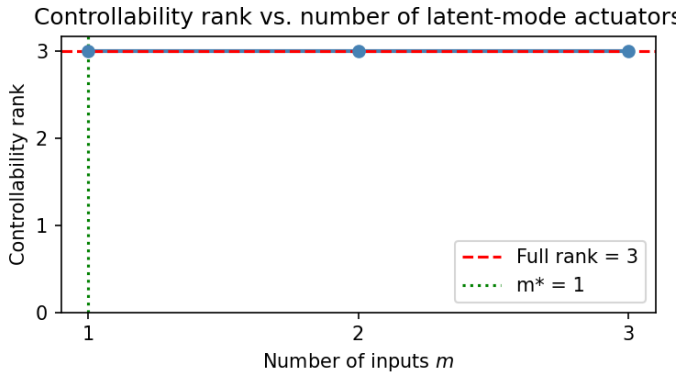

Figure 5: Controllability rank $\operatorname{rank}\mathcal{C}(A_{\mathrm{dmd}}, B)$ as each latent-mode column is added to B in greedy order. The horizontal red dashed line marks full rank $r = 3$; the vertical green dotted line marks the minimum sufficient number of inputs $m^* = 1$.

Figure 6 — Minimum actuator direction in sensor space (`figs/SINDy_B_sensor.png`)

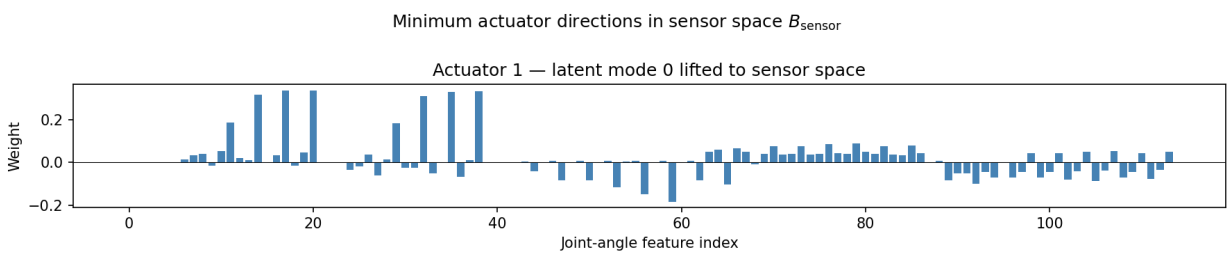

Figure 6: The minimum actuator $B_{\mathrm{sensor}} = U_r[:, 0] \in \mathbb{R}^{114}$, shown as a bar chart over the 114 joint-angle feature indices. Each bar represents the weight of that feature in the optimal actuator direction.

What it tells us. The actuator is sparse in sensor space: only about 8 features (joint-angle channels) carry significant weight ($|w| > 0.1$), while the remaining ≈ 106 features are near zero. The dominant features — indices $\{11, 14, 17, 20, 32, 35, 38\}$ with weights ≈ 0.19–0.34 — correspond to joints that differ most systematically between jumping, running, and walking (e.g. hip, knee, and ankle joints in the frontal and sagittal planes). Physically, B_{sensor} is the actuator pattern that most efficiently shifts the latent state: applying a unit force along B_{sensor} maximally excites the first MTDMD mode, which encodes the inter-class posture difference.

6 Analysis and Discussion

6.1 Why is R^2 only 13.6%?

The low SINDy R^2 is *not* a failure of the method; it is a **physically meaningful result**. The decomposition $\dot{x} = f(x) + Bu$ splits the observed derivative into an autonomous part and a driven part. Our finding that $f(x)$ explains only 14% of $\dot{z}$ implies that approximately 86% of the observed state change at each timestep is attributable to the control input $u(t)$ — the muscle forces and ground reaction forces driving the motion.

This makes intuitive sense: human motion is highly *actuated*. Without active muscle control, a standing human falls. The autonomous dynamics f capture the slow background drift (gravity-induced settling, passive joint compliance), while the bulk of the motion — the deliberate stride, the jump trajectory — is controlled.

A secondary reason for the low R^2 is model mismatch: the degree-2 polynomial library may not be the ideal basis for oscillatory biomechanical dynamics. A Fourier-type library (trigonometric terms) would be better suited but would require more data (many gait cycles) for reliable sparse identification.

6.2 Comparison of A_{sindy} and $A_{\text{dmd}} - I$

In the continuous-time limit ($\Delta t \to 0$), the discrete operator $A_{\text{dmd}} \approx I + \Delta t\, L$ where L is the continuous generator. SINDy approximates $\dot{z} \approx Lz$ with the same $\Delta t = 1$, so ideally $A_{\text{sindy}} \approx A_{\text{dmd}} - I$.

From (6), the entries of $A_{\text{dmd}} - I$ are of order 10^{-3}, while A_{sindy} has entries of order 10^{-2}—one order of magnitude larger. This discrepancy occurs because the STLSQ threshold (5×10^{-3}) kills the small true linear coefficients ($\sim 10^{-3}$) and replaces them with slightly larger spurious values estimated from the correlated quadratic terms. At finer thresholds ($\sim 10^{-4}$) the equations become dense and numerically unreliable. This is the fundamental tension in SINDy for near-identity systems: the true linear dynamics are too small relative to the noise floor.

6.3 Why does one actuator suffice?

A classical result in linear control theory states that (A, b) is controllable for a single column b if and only if b is not orthogonal to any left eigenvector of A [4]. Equivalently, if A has r distinct eigenvalues, *any* generic vector b makes (A, b) controllable.

Here A_{dmd} has three distinct eigenvalues: $\lambda_1 = 0.99944$ (real) and $\lambda_{2,3} = 0.99974 \pm 0.00491j$ (complex conjugate pair, which counts as a single pair of algebraically distinct values over $\mathbb{R}$). The standard basis vectors e_0, e_1, e_2 are all generic with respect to the eigenvector structure, so each independently achieves full controllability.

Geometric multiplicity argument. More generally, the minimum number of inputs required to controllability is

$$m^* \;\geq\; \max_{\lambda} \text{ geo. mult.}(\lambda, A).$$

Since all geometric multiplicities of A_{dmd} equal 1 (three distinct eigenvalues), $m^* \geq 1$. We have shown constructively that $m^* = 1$ is achievable, giving the tight bound $m^* = 1$.

6.4 Implications for motion synthesis

The result $m^* = 1$ means that — within the 3-dimensional MTDMD latent space — a *single scalar control signal* $u(t) \in \mathbb{R}$ suffices to transition the human body between any two latent states (e.g. from a walking configuration to a running configuration), provided the system is driven by the control input $Bu(t)$ along the direction $B_{\text{sensor}} \approx U_r[:, 0]$.

In practice, "controlling" real human motion means designing a trajectory $u(t)$ that moves $z(t)$ from the walking cluster to the running cluster in finite time. Since A_{dmd} is stable and the three clusters are separated by distances on the order of $\|z_{\text{run}} - z_{\text{walk}}\| \approx 800$ units, a finite-time minimum-energy control can be computed by solving a discrete-time LQR problem in the 3D latent space.

7 Summary Table

Table 4: Summary of all quantitative results.

Quantity	**Value**
Original state dimension n	114
Timesteps per trial T	100
Number of training trials N	15 (5 jumping, 5 running, 5 walking)
MTDMD threshold	97% variance
Retained rank r	3
Energy captured at $r = 3$	99.98%
SINDy library	Degree-2 polynomial, $p = 10$ terms
STLSQ threshold λ	5×10^{-3} (scaled)
Training samples for SINDy	1,485
Non-zero SINDy terms	14 / 30
SINDy R^2 (derivative)	13.6%
A_{dmd} eigenvalues	$0.99944;\ 0.99974 \pm 0.00491j$
Minimum inputs m^*	**1**
Actuated latent mode	0 (or any of $\{0, 1, 2\}$)
B_{sensor} non-negligible features	≈ 8 out of 114

References

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[3] R. E. Kalman, "On the general theory of control systems," *IFAC Proceedings Volumes*, 1(1):491–502, 1960.

[4] J. P. Hespanha, *Linear Systems Theory*, 2nd ed. Princeton University Press, 2018.